Untitled

Stop Babysitting Spreadsheets. Start Commanding Data

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The Excel Refugee Migration

Dedication

For the people who kept the reporting cycle alive long after the workbook should have died.

About This Book

This book is for professionals who never planned to become technical people and still ended up carrying technical work anyway.

You might work in finance. Or HR. Or operations. Or reporting. Or business analysis. The exact title on your email signature matters less than the shape of your week. Files arrive. Formats drift. Lookups break. A dashboard needs updating before a meeting. Someone upstream changes a column name and half your morning disappears.

You know the work.

You know the pressure.

You know the strange fatigue of doing something familiar over and over and still not trusting the process.

This book was built for that reader.

Not the aspiring software engineer.

Not the person who wants to learn programming as a hobby.

The working professional who is smart, capable, overloaded, and tired of babysitting spreadsheets.

Who This Book Is For

This book is written for people whose work depends on data but whose jobs were never designed to turn them into full-time technical specialists.

You are likely a fit for this book if you:

- work daily in Excel
- handle recurring reports, reconciliations, merges, or dashboards
- spend too much time cleaning exports and fixing broken workbook logic
- feel comfortable with business reasoning but not with programmer culture

- want results fast, not months of technical theory
- need something useful for Monday morning, not "someday"

You do not need a developer identity to benefit from this book.

You need a better workflow.

What This Book Is Trying to Solve

At a certain point, spreadsheet work stops feeling like analysis and starts feeling like maintenance.

That shift is easy to miss because the reports still go out. The numbers still get checked. The file still opens, eventually. From the outside, the process looks functional.

Inside the work, it feels different.

You save backup copies before touching formulas. You keep old versions nearby in case something breaks. You refresh the same logic every week and still do not fully trust it. Hours disappear into cleaning, checking, copying, pasting, and rebuilding things that should already be stable.

The problem is not that you are disorganized.

The problem is not that you are bad with data.

The problem is that your work has outgrown the shape of the tool.

This book is a response to that moment.

How to Read This Book

Do not read this like a programming manual.

Read it like a migration plan.

Each chapter moves from a familiar business pain point toward a more stable way of working. The story matters because burnout is part of the problem. The code matters because relief has to become operational.

You will get the most from this book if you use it in three passes.

First, read for recognition. Let the book name the patterns you have been carrying.

Second, read for translation. Pay attention to how familiar Excel habits become R workflows.

Third, read for transfer. Use the chapter exercises and code examples on a real reporting task from your own work.

You do not need to master everything at once.

You need one working win, then another.

What You Will Need

You do not need a computer science background.

You do not need to become a programmer.

You do need a few practical things:

- a laptop with R and RStudio installed
- a few sample CSV or Excel files from work, with sensitive information removed if needed
- the willingness to stop treating manual repetition as normal
- enough honesty to map where your time is really going

That is enough to begin.

What You Will Learn to Do

By the end of this book, you should be able to:

- import real business data into R without panic
- clean messy datasets with confidence
- merge large files without relying on brittle lookup formulas
- replace repetitive spreadsheet labor with reusable workflows
- create grouped summaries and charts more efficiently
- use ChatGPT as a translator between Excel logic and R code
- save hours every week by reducing manual reporting work

The focus throughout is practical usefulness.

This is not a book about becoming technically impressive.

It is a book about becoming operationally dangerous in the best possible way.

What Makes This Book Different

Many technical books start too far from the problem.

They begin with syntax, abstract concepts, or a culture of expertise that assumes the reader wants to become part of that world. This book starts somewhere else.

It starts where corporate professionals usually live.

Late files. Dirty exports. Conflicting versions. Hidden columns. Broken formulas. Reports that need to be right before someone more senior walks into a room.

The transition into R is framed here as a professional upgrade, not a personality transplant.

You are not abandoning your existing strengths.

You are extending them into a workflow that scales better.

Book Roadmap

The structure follows a professional transition arc.

- First, you recognize the spreadsheet ceiling and name the real problem.
- Then, you see why R is often a better fit for this kind of work than people expect.
- Then, you build confidence through small practical wins.
- Then, you learn to translate familiar Excel thinking into repeatable R logic.
- Then, you begin using automation, joins, summaries, and charts in ways that save real time.
- Finally, you move from spreadsheet survivor to R-powered strategist.

This is not a straight line toward technical purity.

It is a useful line toward control.

Optional Setup Check

The code blocks throughout the book are meant to be executable. This quick check confirms that your R session is working.

#| label: setup-check #| eval: true

version$version.string

Introduction: The Day My Spreadsheet Died

Chapter Opening Story

You know the report before you open it.
Not the numbers. The ritual.
The workbook lives in a folder called something like `Weekly Reporting Final`, inside another folder called `Final Reports`, beside three other versions with names you no longer trust: `v2`, `final_updated`, `use_this_one`. Every Monday morning, before the first meeting, before the inbox fully wakes up, you open the same file and begin the same sequence.
By now your hands move ahead of your thoughts.
Open the export from finance. Copy the latest transaction rows. Paste values into the raw data tab. Wait for the screen to pause. Switch to the lookup sheet. Check whether last week's formulas still extend far enough. They usually do not. Drag them down another few thousand rows. Watch the formula bar fill with nested logic you once understood perfectly and now read more by instinct than by comprehension.
A cup of coffee sits beside the keyboard, gone cold long enough to lose its smell.
There are eight tabs open. Maybe ten.
HR headcount on one screen. Payroll extract on another. Sales snapshot in a separate workbook because the original file slows to a crawl if everything stays open at once.
Somewhere in the background, email notifications continue arriving, small flashes of demand at the edge of your concentration.
You ignore them.
First, fix the dates.
For reasons nobody ever explains, one file exports dates as text and another as serial numbers. You correct them the way you always do. Quick formula. Copy down. Paste as values. Remove the helper column. Repeat.
Then the duplicates.
There are always duplicates.
Sometimes because the system exported amended records twice. Sometimes because someone upstream merged two extracts manually before sending them over. Sometimes because no one really knows where the clean source lives anymore.
The work is not difficult. That is what makes the exhaustion harder to explain.
You are not confused.
You are tired.
There is a difference.
You know every hidden column in this workbook. You know which tab must never be touched because one broken reference can ripple through the entire file. You know the brittle chain of `VLOOKUP`s holding together data that was never meant to live in a spreadsheet in the first place.
And still every week feels like starting over.
This is the part no one sees.
From the outside, the finished report looks polished. Leadership sees a clean summary, a few directional comments, a chart that seems straightforward enough. It looks controlled. It might even look easy.
They do not see the invisible hour spent tracing a broken formula through six tabs because someone inserted a column upstream.
They do not see the backup copy you save before making changes.
They do not see the strange little dread that arrives when the screen freezes for a second too long.
Late in the cycle, often in silence, the thought arrives with exhausted clarity.

Why This Book Exists

This book exists for professionals who have quietly become the infrastructure.

The finance analyst who spends Monday mornings stitching together exports from three systems.

The HR reporting lead cleaning employee records and headcount tables every week.

The operations manager rebuilding the same dashboard by hand because the source files never arrive in the same format twice.

The business analyst merging sales, forecast, and actuals data through formula chains that feel one edit away from collapse.

If that is you, the first thing to understand is simple.

You are not bad at your job.

In many cases, the opposite is true.

You have become exceptionally good at forcing inadequate tools to behave like systems. You know how to rescue broken workbooks. You know how to trace weird row mismatches. You know when a pivot table is lying. You know the difference between a clean export and a file that will cost you an hour before you even begin.

That competence is real.

The fatigue is real too.

Excel is not the villain here. It is one of the most useful business tools ever made. It gives smart professionals a fast way to inspect, organize, calculate, and communicate. The problem starts later, when a flexible analysis tool becomes the permanent operating system for recurring, large-scale workflows.

A quick fix hardens into a routine.

A one-off workbook becomes a weekly reporting dependency.

A helper column becomes a logic chain nobody wants to touch.

A file built for convenience becomes a fragile process people are afraid to change.

And because the work still gets done, people assume the process still works.

It does not really work.

It survives.

There is a cost to that survival. High-quality attention gets spent on low-value repetition. Your judgment gets crowded out by maintenance. Instead of thinking about what the data means, you spend your best hours preserving the machinery around it.

That is why this book exists.

Not to turn you into someone else.

Not to push you into programmer culture.

Not to tell you that everything you know is wrong.

This book exists to help you keep what is strongest in the way you already think, then move that intelligence into a workflow that scales better.

You do not need a new personality.

You need a better system.

What This Book Will Help You Do

By the end of this book, you will be able to take work that currently eats hours every week and turn it into processes you trust.

You will know how to import real business files into R and inspect them without panic. You will know how to clean messy exports without fixing cells one by one. You will know how to merge large datasets without relying on brittle lookups. You will know how to automate recurring reporting logic so the process becomes repeatable instead of fragile.

The emphasis here is speed to result.

This is not a theory-first book. It is built around Monday morning work. A payroll reconciliation. A sales performance extract. A headcount merge. A weekly KPI pack. A chart for leadership that needs updated numbers before 10 a.m.

Each chapter is designed to move from a recognizable business problem toward a usable outcome.

That matters because many professionals do not need more information. They need relief that becomes practical fast.

R is introduced here as the natural next step for people who already think in tables, business rules, summaries, exceptions, and reporting logic. You already have the instincts this work requires. You already know how to reason with data. What changes is the environment in which that reasoning happens.

Instead of dragging formulas across thousands of rows, you will learn to work at the column level.

Instead of rebuilding the same logic every week, you will learn to run repeatable workflows.

Instead of holding a process together with memory and vigilance, you will learn to give the process structure.

The emotional payoff is straightforward.

You save hours every week.

The professional payoff runs deeper than that. Less fatigue. Fewer silent errors. More confidence in your outputs. More room to think. More trust in the work you hand over. More credibility when the conversation shifts from “Can you pull the numbers?” to “What do you see here?”

That is the real movement of this book.

From spreadsheet survival to analytical control.

PhD Insight Made Simple

Manual spreadsheet workflows create hidden cognitive load.
Every repeated copy-paste, formula drag, format correction, and reconciliation step consumes decision energy. Even when the task feels familiar, your brain is still making small operational judgments. Which file is current. Did the formulas extend. Was the pivot refreshed. Did the row count change. That constant checking creates decision fatigue, even when the work looks routine from the outside.
Repeatable workflows reduce that burden by shifting effort from memory to system. Once a process is scripted and reusable, the workflow carries the logic for you. You stop rebuilding the same path every week and start using your attention where it matters most: review, interpretation, and judgment.

Monday Morning Application

Choose one report or spreadsheet workflow you repeat most often at work.
Write down the answers to these five questions:

1. What is the name or purpose of the report?
2. How long does it take from raw file to final output?
3. Which steps are manual each time?
4. What breaks most often: formulas, formats, lookups, merges, or refresh logic?
5. Which parts do you copy, paste, or rebuild every week?

Do not optimize anything yet.
Do not try to solve it on the spot.
Just map the workflow honestly. This becomes your baseline, and later it becomes your proof that the old way was costing more than it seemed.

Main

The Spreadsheet Ceiling: Why Your Excel Skills Became a Prison

Chapter Opening Story

The workbook does not have to crash for you to know something is wrong.
Sometimes the signal is quieter than that.
It shows up in the way you hesitate before opening a file you know too well. In the extra backup copy you save before touching a lookup tab. In the private relief you feel when a meeting gets moved because the report was not going to be ready anyway. Burnout in spreadsheet work rarely looks dramatic. It looks organized. It looks competent. It looks like somebody who still delivers, but at a growing internal cost.
That is why the ceiling is hard to name.
From the outside, you still seem effective. Deadlines are met. Reports go out. Numbers get checked. But behind the scenes, the process has become heavier than it appears. More manual. More fragile. Less trustworthy. You are no longer using Excel as a tool. You are holding together a system it was never meant to carry.

The Cost of Holding a Workflow Together by Hand

A surprising amount of corporate work now depends on individual stamina.

Not documented systems. Not stable pipelines. Stamina.

One person knows which file is the source truth this week. One person remembers that the upstream export changed its date format last month. One person notices when the row counts do not reconcile. One person understands why the pivot table looks right and still is not.

That person is often you.

The cost of this arrangement is easy to underestimate because each manual step feels small on its own. Copy a tab. Extend a formula. Rename a few fields. Remove blanks. Refresh the pivot. Paste values. Save another version. None of those steps looks serious in isolation.

Put them together, repeat them weekly, and you get a job shaped around maintenance.

That shift matters. Maintenance work is sticky. It eats high-quality attention. It creates low-grade anxiety because every repeated action also carries a failure risk. You can be excellent at the work and still feel increasingly trapped by it.

When Familiar Tools Become Operational Risk

Familiarity feels safe until it starts hiding the damage.

The more fluent you become in Excel rescue work, the easier it is for people around you to assume the process is healthy. It is not healthy because it functions. It functions because you compensate for its weaknesses in real time.

That compensation has a cost.

When a process depends on memory, patchwork fixes, and silent expertise, it becomes harder to delegate, harder to audit, and harder to improve. Small changes upstream ripple downstream. An inserted column breaks a lookup. A renamed file breaks a macro. A duplicated extract distorts a summary. And because the errors often appear late, after hours of work, the emotional cost compounds with the technical one.

The spreadsheet ceiling is not simply about file size. It is about the point where the workflow asks more from you than the tool can reliably support.

The Emotional Tax of Fragile Workbooks

There is a specific kind of tiredness that comes from not trusting your process.

You check things twice. Then again. You compare totals manually because you know a silent mismatch is possible. You keep old versions nearby in case the current one breaks. You postpone changes because the system is too brittle to tolerate experimentation.

This is one reason smart professionals start to feel smaller inside spreadsheet-heavy jobs. Their intelligence is still there. Their judgment is still there. But more and more of it gets redirected into guarding against avoidable failure.

That is not what analytical work is supposed to feel like.

The First Honest Question

A useful turning point begins with a question many professionals delay longer than they should.

What if the real problem is not your discipline, but the structure of the workflow itself?

That question creates room for a different kind of solution.

Not another patch.

Not a prettier workbook.

A change in how the work is done.

Cheat Sheet Box

Signs you have hit the spreadsheet ceiling

- one report has multiple "final" versions
- formula chains are too fragile to modify confidently
- weekly refreshes depend on memory rather than a documented process
- large files slow down basic tasks
- you spend more time preparing data than interpreting it

PhD Insight Made Simple

A workflow ceiling happens when a process still produces output but only through increasing effort, vigilance, and workaround behavior. The work appears stable on the surface, but the effort required to keep it stable keeps rising. That is a structural limit, not a personal failure.

Visual Breakout

Create a visual with two columns.
On the left, show a single workbook with arrows coming into it from payroll, CRM, HR, finance, and manual notes.
On the right, show a pipeline with steps: import, clean, join, summarize, visualize, export.
The contrast should make one point clear: one file is carrying too much.

Monday Morning Application

Take one recurring report and list every manual step from file arrival to final output. Then circle the steps that rely on memory, copy-paste, manual checking, or fragile formulas.
Those circles mark the first targets for migration.

Optional Workflow Snapshot

```r
library(tibble)

workflow_steps <- tibble::tribble(
  ~step, ~manual, ~common_failure,
  "Import raw file", TRUE, "Wrong version selected",
  "Extend formulas", TRUE, "Formula range stops early",
  "Refresh pivot", TRUE, "Pivot points to old range",
  "Create chart", TRUE, "Labels out of date"
)
```

```
workflow_steps
```

Why Not Python? The Honest Answer Nobody Gives You

Chapter Opening Story

You may already have tried to solve this problem the respectable way.
You opened a few Python tutorials. Watched someone build something elegant in a terminal window. Tried to follow along. Made it halfway through variables, loops, environments, maybe notebooks, maybe package installs, before a quiet resistance set in.
Not because you are incapable.
Because the route felt strangely far away from the kind of work you do.
You were not trying to build software. You were trying to clean a payroll extract without losing your afternoon. You were trying to merge sales actuals with forecast data. You were trying to update a report people needed in two hours. The distance between your business problem and the learning path being offered felt wider than it should have been.
That distance matters.
A lot of professionals decide they are "not technical" at exactly that point, when the truth is simpler. They met a tool through the wrong door.

The Wrong First Tool for the Wrong Kind of Fear

Fear around programming is often misdiagnosed.

People assume the problem is discipline, intelligence, or confidence. Often it is none of those. Often the problem is mismatch. The learner needs speed, relevance, and visible business payoff. The learning path offers abstractions first and usefulness later.

That is a hard bargain for busy professionals.

When your days are already crowded, you do not want to study a language in the abstract. You want to reduce friction in work you already do. The right entry point matters because early wins shape whether the transition feels possible.

Why R Feels Closer to Spreadsheet Logic

R makes sense to many Excel-heavy professionals because it meets them near the shape of their existing thinking.

You already think in tables.

You already think in rows, columns, filters, summaries, categories, exceptions, and business rules.

You already understand the logic of joining related information, even if you have been doing it through `VLOOKUP`, `INDEX` and `MATCH`, or `XLOOKUP`.

R is not identical to Excel. It should not be. But for many analysts, it feels closer to the work itself. It is built around data workflows. Import the file. Clean the names. Filter the rows. Group the values. Join the tables. Build the chart. Export the result.

That progression feels legible.

Business Workflows Need Fast Wins

In a corporate setting, usefulness wins.

Not elegance. Not purity. Usefulness.

A good first tool should help you solve recognizable problems quickly. That is where R often becomes compelling. You can move from raw CSV to grouped summary in a short stretch of code. You can clean one messy export and feel the difference immediately. You can replace a brittle manual sequence with something stable enough to trust.

That speed changes the emotional experience of learning.

You stop feeling like an outsider visiting somebody else's technical world. You start feeling like a professional extending your existing capabilities.

You Are Not Becoming a Developer

This matters enough to say clearly.

You do not need to adopt a developer identity to benefit from R.

You can remain what you already are: a finance professional, an HR analyst, an operations lead, a reporting specialist, a business analyst.

What changes is not your professional worth. What changes is the leverage behind your work.

Cheat Sheet Box

Excel mindset vs R workflow mindset

Excel habit	R equivalent mindset
Fix cell by cell	Transform column by column
Search workbook manually	Filter and inspect programmatically
Lookup through formulas	Join related tables
Refresh pivot by hand	Recompute summary through code
Save new version each week	Re-run the same script

PhD Insight Made Simple

Tool fit matters in adult learning. When a tool maps closely to the problems a learner already solves, cognitive friction drops. Less effort is spent translating the task into the tool. More effort goes straight into useful output. That is one reason some analysts find R easier to adopt than more general-purpose languages.

Visual Breakout

Create a three-column comparison.
Column 1: Excel, labeled familiar but fragile.
Column 2: Python, labeled powerful but broader than needed for many first-time analysts.
Column 3: R, labeled analyst-friendly bridge.
Use rows for learning friction, speed to first result, fit for reporting, and comfort with table-based work.

Monday Morning Application

Write down one workflow you want to speed up in the next two weeks.
Then answer two questions:

1. Does this workflow mostly involve tables, cleaning, merging, summarizing, or charting?
2. Would a repeatable script solve more of the pain than a better spreadsheet?

If the answer to both is yes, you are already describing an R-shaped problem.

First Taste of the Workflow

```
library(readr)
library(dplyr)
```

```r
sales <- read_csv("data/sales_export.csv")

sales |>
  filter(region == "West") |>
  group_by(product_category) |>
  summarise(
    revenue = sum(revenue, na.rm = TRUE),
    orders = n(),
    .groups = "drop"
  )
```

R Without the Fear: Your First 60 Minutes (And It Won't Hurt)

> **Chapter Opening Story**
>
> Placeholder for the chapter-opening story. Start with a reader expecting technical pain and finding relief through one small, usable win.

What You Need Installed and Nothing More

Placeholder starter prose.

```
library(readr)
library(dplyr)
```

The First Dataset

Placeholder starter prose.

Rows, Columns, and Filters Without Formula Anxiety

Placeholder starter prose.

Save Something Useful in the First Hour

Placeholder starter prose.

> **PhD Insight Made Simple**
>
> Placeholder for data frame explanation in plain language.

Visual Breakout

Placeholder visual concept: Excel table vs R data frame.

Monday Morning Application

Import one real work file and filter it by a business condition.

Fluent in Both Languages: Translating Your Excel Brain to R

Chapter Opening Story

Placeholder for a recognition scene where the reader realizes R follows business logic they already understand.

From Cells to Columns

Placeholder starter prose.

From Dragging Formulas to Column-Level Logic

Placeholder starter prose.

From Tabs and Files to Structured Tables

Placeholder starter prose.

From Manual Refresh to Repeatable Script

Placeholder starter prose.

Cheat Sheet Box

Excel logic	R logic
`VLOOKUP`	`left_join()`
Pivot table	`group_by()` + `summarise()`
`IF()`	`case_when()`
Filter	`filter()`

PhD Insight Made Simple

Placeholder for vectorized operations explained simply.

Visual Breakout

Placeholder visual concept map: Excel logic to R logic.

Monday Morning Application

Translate one familiar Excel workflow into a short R sequence.

Your AI Co-Pilot: Using ChatGPT as Your Personal R Translator

> **Chapter Opening Story**
>
> Placeholder for a scene where the reader gets unstuck by asking for translation instead of memorizing syntax.

Why Translation Beats Memorization

Placeholder starter prose.

Turning Excel Formulas Into R Code

Placeholder starter prose.

Turning VLOOKUP Thinking Into Joins

Placeholder starter prose.

Turning Pivot Table Thinking Into Summaries

Placeholder starter prose.

Prompting ChatGPT Without Getting Lost

Placeholder starter prose.

How to Verify Outputs Before Sending Anything Upstream

Placeholder starter prose.

Cheat Sheet Box

Prompt template placeholder.

PhD Insight Made Simple

Placeholder for human-in-the-loop verification.

Visual Breakout

Placeholder visual: Excel problem -> prompt -> R draft -> verification -> output.

Monday Morning Application

Paste one live Excel formula into ChatGPT and ask for an R translation.

```r
# Example verification workflow placeholder
result_rows <- 1250
expected_rows <- 1250

identical(result_rows, expected_rows)
```

Clean Data, Clean Mind: The R Workflow That Eliminates Manual Hell

> **Chapter Opening Story**
>
> Placeholder for a dirty-file scene from HR, finance, or operations.

Why Dirty Data Creates Dirty Thinking

Placeholder starter prose.

The Repeatable Cleaning Sequence

Placeholder starter prose.

Duplicates, Blanks, Dates, and Strange Text

Placeholder starter prose.

Saving the Clean Version Without Starting Over Every Week

Placeholder starter prose.

```r
library(readr)
library(dplyr)
library(stringr)
library(janitor)

raw_data <- read_csv("data/raw_export.csv")

clean_data <- raw_data |>
  janitor::clean_names() |>
```

```
mutate(across(where(is.character), str_trim)) |>
distinct()
```

PhD Insight Made Simple

Placeholder for tidy data principles.

Visual Breakout

Placeholder visual: dirty file to clean table.

Monday Morning Application

Clean one messy exported file from work.

Pivot Tables Are Dead. Long Live dplyr.

> **Chapter Opening Story**
>
> Placeholder for a reporting task involving repeated pivot rebuilds.

What Pivot Tables Get Right

Placeholder starter prose.

Where Pivot Tables Run Out of Road

Placeholder starter prose.

Grouped Summaries That Stay Consistent

Placeholder starter prose.

From One-Off Report to Reusable Workflow

Placeholder starter prose.

```r
library(dplyr)

sales |>
  group_by(region, month) |>
  summarise(
    revenue = sum(revenue, na.rm = TRUE),
    orders = n(),
    .groups = "drop"
  )
```

Cheat Sheet Box

Placeholder comparison: pivot fields vs `dplyr` verbs.

PhD Insight Made Simple

Placeholder for grouped aggregation.

Visual Breakout

Placeholder visual: Pivot Table vs `group_by()` and `summarise()`.

Monday Morning Application

Recreate one recurring pivot summary in R.

Charts That Actually Impress: ggplot2 vs. Excel's Chart Wizard

> **Chapter Opening Story**
>
> Placeholder for a before-and-after chart scene where the visual quality changes the conversation.

Why So Many Spreadsheet Charts Look Unfinished

Placeholder starter prose.

What Makes a Leadership-Ready Chart

Placeholder starter prose.

Your First Good Plot in ggplot2

Placeholder starter prose.

From Pretty Enough to Decision-Ready

Placeholder starter prose.

```r
library(ggplot2)

ggplot(sales_summary, aes(x = month, y = revenue)) +
  geom_line() +
  labs(
    title = "Monthly Revenue Trend",
    x = NULL,
    y = "Revenue"
  )
```

PhD Insight Made Simple

Placeholder for visual encoding in plain language.

Visual Breakout

Placeholder visual: Excel default chart vs refined ggplot2 output.

Monday Morning Application

Build one weekly chart for a real stakeholder.

Your First Real Project: A Full Analysis From Raw File to Boardroom Slide

> **Chapter Opening Story**
>
> Placeholder for a realistic cross-functional reporting request with multiple files and a decision deadline.

The Business Question

Placeholder starter prose.

The Raw Files Arrive

Placeholder starter prose.

Clean, Merge, Summarize

Placeholder starter prose.

Build the Slide They Will Remember

Placeholder starter prose.

What This Project Proves About You

Placeholder starter prose.

```r
library(readr)
library(dplyr)

headcount <- read_csv("data/headcount.csv")
payroll <- read_csv("data/payroll.csv")

project_output <- headcount |>
  left_join(payroll, by = "employee_id")
```

Cheat Sheet Box

Placeholder project workflow checklist.

PhD Insight Made Simple

Placeholder for reproducibility.

Visual Breakout

Placeholder pipeline from raw files to final slide.

Monday Morning Application

Reproduce a smaller version of a current work analysis.

The R-Powered Strategist: Your 30-Day Migration Plan and What Comes Next

Chapter Opening Story

Placeholder for a scene where the reader looks back at the old workflow with new confidence.

The Identity Shift

Placeholder starter prose.

Your 30-Day Migration Plan

Placeholder starter prose.

Which Tasks to Move First

Placeholder starter prose.

How to Build Trust With Stakeholders

Placeholder starter prose.

What Comes Next

Placeholder starter prose.

Cheat Sheet Box

Placeholder 30-day migration checklist.

PhD Insight Made Simple

Placeholder for compounding productivity gains.

Visual Breakout

Placeholder visual: spreadsheet operator to R-powered strategist.

Monday Morning Application

Choose one recurring workflow to migrate this month.

Appendix A: The Master Excel-to-R Cheat Sheet

Core Translation Map

Excel term	R equivalent	Plain-language use
Worksheet table	data frame	Structured tabular data
Filter	`filter()`	Keep only rows you need
Sort	`arrange()`	Reorder rows
VLOOKUP/XLOOKUP	`left_join()`	Pull matching columns from another table
Pivot table	`group_by()` + `summarise()`	Build grouped summaries
IF	`if_else()` or `case_when()`	Apply conditional logic
Text cleanup	`stringr`	Clean messy text
Save file	`write_csv()`	Export clean output

Common Patterns

Placeholder for expanded cheat sheet tables and workflow snippets.

Appendix B: The ChatGPT Prompt Library for R

Formula Translation Prompt

> I use Excel for business reporting. Translate this formula into R using tidyverse syntax. Explain each part in plain language and show a small example.

Join Logic Prompt

> I have two tables. One contains employee master data and one contains payroll records. Show me how to merge them in R using employee_id as the key. Also tell me how to check for unmatched rows.

Summary Prompt

> Translate this pivot table goal into R: total revenue by region and month, sorted descending by revenue.

Validation Prompt

> I wrote this R code to replace an Excel report. Tell me how to verify row counts, totals, missing joins, and output quality before I send the report.

Placeholder for expanded prompt library.

Appendix C: Recommended R Packages by Use Case

Data Import

- `readr`
- `readxl`
- `vroom`

Data Cleaning

- `dplyr`
- `janitor`
- `stringr`
- `lubridate`

Data Transformation

- `tidyr`
- `purrr`

Visualization

- `ggplot2`
- `scales`

Reporting

- `quarto`
- `rmarkdown`
- `gt`

File Paths and Project Hygiene

- `here`
- `fs`

Placeholder for fuller package notes and usage examples.

Appendix D: Glossary, Excel Terms and Their R Equivalents

Glossary

Term	Meaning
data frame	A table-like structure used in R for rows and columns of data
join	A way to combine two tables using a shared key
tidy data	A clean data structure where each row is one observation and each column is one variable
script	A saved sequence of instructions you can run again
reproducibility	Getting the same result from the same process
vectorized operation	Applying a rule to a whole column at once rather than cell by cell

Placeholder for expanded glossary.

Author's Note

You did not need another lecture about "learning to code."

You needed relief.

If this book did its job, it gave you more than syntax. It gave you a way to reclaim mental space from maintenance and move toward work that feels sharper, calmer, and more strategic.

Placeholder for final CTA, newsletter, companion workbook, downloadable code pack, or consulting/training link.